

1.

Fisher's linear discriminant is

$$\hat{\mathbf{w}}^* = \arg \max_{\hat{\mathbf{w}}} J(\hat{\mathbf{w}}) = \arg \max_{\hat{\mathbf{w}}} \frac{\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}}$$

where $\mathbf{S}_b = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$ and $\mathbf{S}_w = \sum_j \sum_{\alpha} (\mathbf{x}_{\alpha} - \mathbf{m}_j)(\mathbf{x}_{\alpha} - \mathbf{m}_j)^T$

(a)

By writing $\frac{dJ(\hat{\mathbf{w}})}{d\hat{\mathbf{w}}} = 0$, show that

$$\mathbf{S}_w^{-1} \mathbf{S}_b \hat{\mathbf{w}} = J(\hat{\mathbf{w}}) \hat{\mathbf{w}}.$$

We aim to take the derivative, and setting this to 0 essentially indicates that we're trying to find the maximum of $J(\hat{\mathbf{w}})$ itself. We see from what we are given that

$$J(\hat{\mathbf{w}}) = \frac{\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}}$$

and we take the derivative accordingly:

$$\frac{dJ(\hat{\mathbf{w}})}{d\hat{\mathbf{w}}} = \frac{d}{d\hat{\mathbf{w}}} \left(\frac{\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}} \right)$$

using the quotient rule, so first finding the derivatives of the denominator and numerator by taking the gradient:

$$\frac{d}{d\hat{\mathbf{w}}} (\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}) = 2\mathbf{S}_b \hat{\mathbf{w}}$$

$$\frac{d}{d\hat{\mathbf{w}}} (\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}) = 2\mathbf{S}_w \hat{\mathbf{w}}$$

and plugging back in:

$$\frac{dJ(\hat{\mathbf{w}})}{d\hat{\mathbf{w}}} = \frac{(\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})(2\mathbf{S}_b \hat{\mathbf{w}}) - (\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}})(2\mathbf{S}_w \hat{\mathbf{w}})}{(\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})^2}$$

and solving by setting to 0:

$$0 = \frac{(\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})(2\mathbf{S}_b \hat{\mathbf{w}}) - (\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}})(2\mathbf{S}_w \hat{\mathbf{w}})}{(\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})^2}$$

$$0 = (\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})(2\mathbf{S}_b \hat{\mathbf{w}}) - (\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}})(2\mathbf{S}_w \hat{\mathbf{w}})$$

$$\begin{aligned}
(\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}})(2\mathbf{S}_w \hat{\mathbf{w}}) &= (\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})(2\mathbf{S}_b \hat{\mathbf{w}}) \\
\frac{(\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}})(2\mathbf{S}_w \hat{\mathbf{w}})}{(\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}})} &= (2\mathbf{S}_b \hat{\mathbf{w}}) \\
\frac{\hat{\mathbf{w}}^T \mathbf{S}_b \hat{\mathbf{w}}}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}} \mathbf{S}_w \hat{\mathbf{w}} &= \mathbf{S}_b \hat{\mathbf{w}} \\
J(\hat{\mathbf{w}}) \mathbf{S}_w \hat{\mathbf{w}} &= \mathbf{S}_b \hat{\mathbf{w}} \\
J(\hat{\mathbf{w}}) \hat{\mathbf{w}} &= \mathbf{S}_w^{-1} \mathbf{S}_b \hat{\mathbf{w}} \\
\mathbf{S}_w^{-1} \mathbf{S}_b \hat{\mathbf{w}} &= J(\hat{\mathbf{w}}) \hat{\mathbf{w}}
\end{aligned}$$

exactly as we sought to find.

(b)

Explain why $\hat{\mathbf{w}}^*$ is the eigenvector for which $J(\hat{\mathbf{w}})$ is the maximum eigenvalue of $\mathbf{S}_w^{-1} \mathbf{S}_b$.

As we showed in part (a), we have $\mathbf{S}_w^{-1} \mathbf{S}_b \hat{\mathbf{w}} = J(\hat{\mathbf{w}}) \hat{\mathbf{w}}$. Looking at the standard definition of an eigenvector, with $Ax = \lambda x$, we can map the terms accordingly to see that we have $A = \mathbf{S}_w^{-1} \mathbf{S}_b$ and $\lambda = J(\hat{\mathbf{w}})$, i.e. we have eigenvector x as $\hat{\mathbf{w}}$. The problem statement then defines $\hat{\mathbf{w}}^*$ as the vector that maximizes $J(\hat{\mathbf{w}})$, which we now see is the corresponding eigenvalue to $\mathbf{S}_w^{-1} \mathbf{S}_b$, so we have to be selecting the maximizing corresponding eigenvector here.

(c)

Explain why $\mathbf{S}_b \hat{\mathbf{w}}$ is always in the direction of $\mathbf{m}_1 - \mathbf{m}_2$ and thus show that

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

We were given that $\mathbf{S}_b = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$ – multiply this by $\hat{\mathbf{w}}$:

$$\mathbf{S}_b \hat{\mathbf{w}} = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T \hat{\mathbf{w}}$$

We can break down $(\mathbf{m}_1 - \mathbf{m}_2)^T \hat{\mathbf{w}}$ into the multiplication of a row vector (as the column vectors are subtracted and then transposed) by a column vector, which

results in a 1x1 matrix, which is just a constant. This shows the first part of the statement which we aimed to prove, with $\mathbf{S}_b \hat{\mathbf{w}}$ always being in the direction of $\mathbf{m}_1 - \mathbf{m}_2$ since it multiplies out to a constant multiplied by $\mathbf{m}_1 - \mathbf{m}_2$, which obviously is always in the direction of $\mathbf{m}_1 - \mathbf{m}_2$.

Next, we aim to prove the final equation. Start with one part of our initial derivative computation in part (a):

$$\begin{aligned} J(\hat{\mathbf{w}}) \mathbf{S}_w \hat{\mathbf{w}} &= \mathbf{S}_b \hat{\mathbf{w}} \\ J(\hat{\mathbf{w}}) \mathbf{S}_w \hat{\mathbf{w}} &= (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T \hat{\mathbf{w}} \\ J(\hat{\mathbf{w}}) \mathbf{S}_w \hat{\mathbf{w}} &= \text{const.} \cdot (\mathbf{m}_1 - \mathbf{m}_2) \\ J(\hat{\mathbf{w}}) \hat{\mathbf{w}} &= \text{const.} \cdot \mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2) \\ J(\hat{\mathbf{w}}) \hat{\mathbf{w}} &= \text{const.} \cdot \mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2) \\ \hat{\mathbf{w}} &= \text{const.} \cdot \frac{\mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2)}{J(\hat{\mathbf{w}})} \end{aligned}$$

We can substitute in $\hat{\mathbf{w}}^*$ since this equation must hold for all points, this must also hold for the most optimal $\hat{\mathbf{w}}$:

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \frac{\mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2)}{J(\hat{\mathbf{w}}^*)}$$

We can also see that $J(\hat{\mathbf{w}}^*)$ here must be a scalar as an eigenvalue, so this can also be treated as a constant, leaving us with

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

which is exactly what we sought to find.

2.

Another way to optimize Fisher's linear discriminant (suggested by Barry Fridling):

(a)

Show that for any two real vectors $\mathbf{x}$ and $\mathbf{y}$

$$(\mathbf{x}^T \mathbf{y})^2 \leq (\mathbf{x}^T \mathbf{x})(\mathbf{y}^T \mathbf{y}), \quad (\text{Cacuhy-Schwarz})$$

(b)

Show that if the λ_k are positive,

$$\left(\sum_{k=1}^N x_k y_k \right)^2 \leq \left(\sum_{k=1}^N \lambda_k x_k^2 \right) \left(\sum_{k=1}^N y_k^2 / \lambda_k \right).$$

(c)

Thus, show that for $\mathbf{A}$ positive definite

$$(\mathbf{x}^T \mathbf{y})^2 \leq (\mathbf{x}^T \mathbf{A} \mathbf{x})(\mathbf{y}^T \mathbf{A}^{-1} \mathbf{y}).$$

(d)

By letting $\mathbf{A} = \mathbf{S}_w$ in the expression above, and writing

$$J(\hat{\mathbf{w}}) = \frac{|\hat{\mathbf{w}}^T (\mathbf{m}_1 - \mathbf{m}_2)|^2}{\hat{\mathbf{w}}^T \mathbf{S}_w \hat{\mathbf{w}}}$$

show again that

$$\hat{\mathbf{w}}^* = \text{const.} \cdot \mathbf{S}_w^{-1}(\mathbf{m}_1 - \mathbf{m}_2).$$

3.

Using the Optdigits dataset from the UCI repository, implement PCA. Reconstruct the digit images and calculate the error $E(n) = \sum_{j=1}^m \|\hat{\mathbf{x}}_{j,n} - \mathbf{x}_j\|^2$ for various values of n , the number of principal components included in the reconstruction of each image. Plot $E(n)$ versus n .